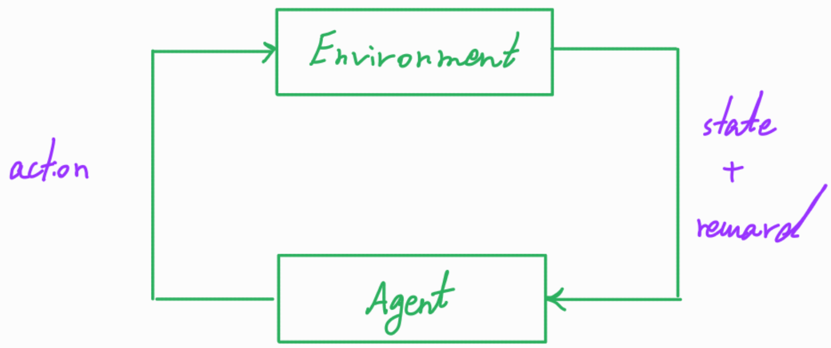
ECE 59500: Reinforcement Learning - Theory & Algorithms

Reinforcement Learning (RL): A set of interrelated problems of learning from interaction that fall in the category of sequential decision-making under uncertainty.

Agent: The decision-maker, e.g., an autonomous system, that decides about the actions.

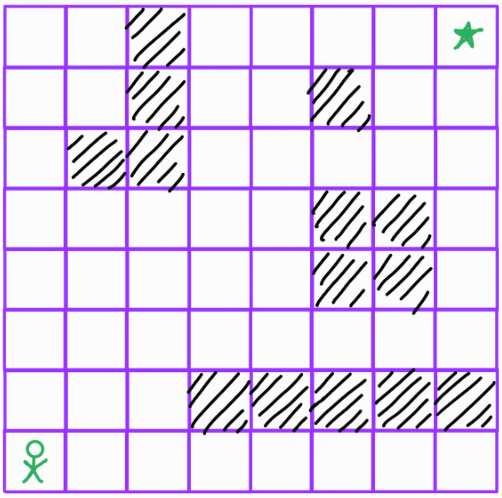
Environment: Everything in the world of the agent.



Key aspects of RL:

- closed-loop interaction
- no direct supervision
- Long-term consequence of ctions

Example: Navigation in a 2D grid-world



Agent $\longrightarrow$ robot

Environment $\longrightarrow$ grid-world

state $\longrightarrow$ $s \in \{1, 2, 3, \dots, 64\}$

$\longrightarrow$ $s \in \{(x, y) \mid x, y \in \{1, 2, \dots, 8\}\}$

$\longrightarrow$ $s \in \{(\Delta x, \Delta y) \mid \Delta x, \Delta y \in \{-7, -6, \dots, 6, 7\}\}$

$\longrightarrow$ $s \in \mathbb{R}^{128 \times 128}$ 128x128 grayscale image

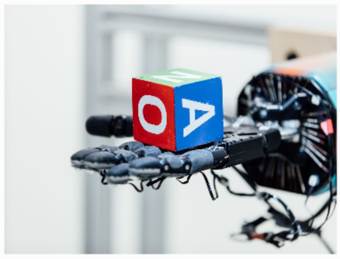
$\longrightarrow$ $s \in \{(loc_{robot}, loc_{obs}, loc_{target})\}$ for moving obstacles and targets

Action $\longrightarrow$ $a \in \{\text{'up'}, \text{'down'}, \text{'left'}, \text{'right'}, \text{'stop'}\}$

Reward $\longrightarrow$ $r = \mathbb{1}[s_{robot} = s_{target}]$ sparse reward

$\longrightarrow$ $r = -\text{dist}(\text{robot}, \text{target}) \underbrace{+ \text{dist}(\text{robot}, \text{nearest object})}_{\text{for obstacle avoidance}}$

Applications of RL:



- Games $\longrightarrow$ Backgammon (TD-Gammon), chess (Alpha Zero), Go (AlphaGo), Dota 2 (OpenAI Five), interesting NPCs
- Robotics $\longrightarrow$ open AI dexterous manipulation, autonomous driving

- Marketing $\longrightarrow$ recommendation systems
- Conversational AI $\longrightarrow$ chat GPT

- Network Design and Control
- Combinatorial optimization

RL connection to control theory

Optimal control : Design a controller to minimize a measure of a dynamical system's behavior over time

Adaptive optimal control : Optimal control with unknown dynamics

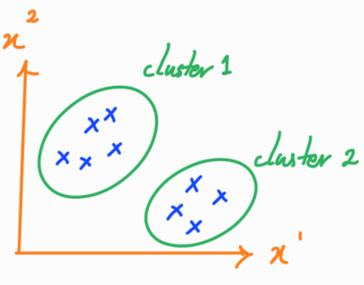
$\underbrace{\text{Adaptive optimal control}}_{\equiv \text{ Model-based RL}}$

- Hamilton - Jacobi - Bellman (HJB) equation $\rightarrow$ continuous-time system
- Bellman equation $\rightarrow$ discret-time system

RL connection to machine learning

Machine Learning $\rightarrow$

- unsupervised learning $\{x_i\}_{i=1}^{N}$ — descriptive analytics — clustering, dimensionality reduction
- supervised Learning $\{(x_i, y_i)\}_{i=1}^{N}$ — predictive analytics — regression, classification
- reinforcement learning $\{(s_t, a_t, r_t)\}_{t=1}^{T}$ — prescriptive analytics



clustering

$x_i = (x_i^1, x_i^2)$

classification

$x_i = (x_i^1, x_i^2)$

y_i = ★ or •

Topics under or related to RL

- Bandit algorithms
- RL in partially observable settings
- Risk-aware RL
- Imitation learning
- RL from human feedback
- Multi-agent RL
- Multi-task learning, meta learning, and transfer learning
- RL with side information